EXPERIMENT 2

Perform recursive binary search using C.

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Class:-SY1 Batch:-B1 PRN:B25ET1046 Div:-1

Input:-

```
#include <stdio.h>
int BS(int arr[], int l, int r, int target);

int main()
{
    int arr[] = {7, 22, 30, 45, 76};
    int n = 5;
    int target;

    printf("Enter the target: ");
    scanf("%d", &target);

    int result = BS(arr, 0, n - 1, target);

    if (result == 1)
        printf("The target is found.\n");
    else
        printf("The target is not found.\n");

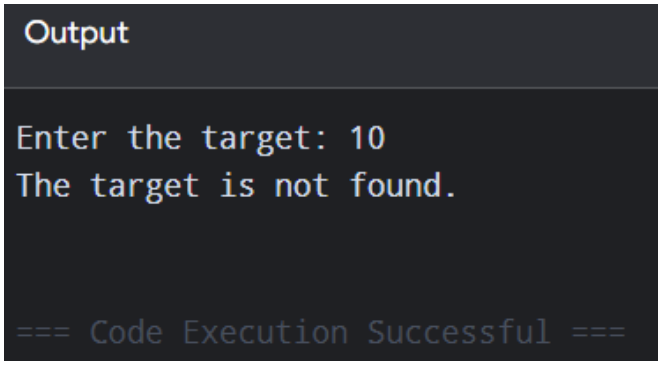
    return 0;
}

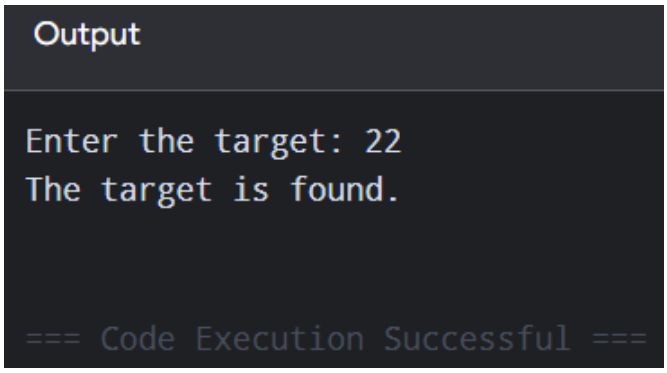

int BS(int arr[], int l, int r, int target)
{
    if (l > r)
        return 0;

    int mid = (l + r) / 2;

    if (arr[mid] == target)
        return 1;
    else if (arr[mid] > target)
        return BS(arr, l, mid - 1, target);
    else
        return BS(arr, mid + 1, r, target);
}
```

Output:-


Output
Enter the target: 10
The target is not found.
=== Code Execution Successful ===


Output
Enter the target: 22
The target is found.
=== Code Execution Successful ===